MA378: Tutorial Sheet 5

All questions are from old exam papers; some likely featured on other Tutorial Sheets

Also: Let p_n be the polynomial of degree n that interpolates f at the $n+1$ points $a = x_0 < x_1 < \cdots < x_n = b$. You may assume that, for any $x \in [a, b]$ there is a $\tau \in (a, b)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!}\pi_{n+1}(x), \tag{1}$$

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Q1. (Q2, 24/25)

(a) Let $f(x) = xe^x$. Calculate the Lagrange form of p_2, the polynomial interpolant to f at the interpolation points $\{x_0, x_1, x_2\} = \{0, 1, 2\}$.

(b) Using Equation (1), find an upper bound for $|f(x) - p_2(x)|$, at $x = 1.5$.

(c) Let p_n be the polynomial of degree n that interpolates an arbitrary function f at the $n+1$ equidistant points $-1 = x_0 < x_1 < \cdots < x_n = 1$.
Can one expect that $\max_{-1 \le x \le 1} |f(x) - p_n(x)| \to 0$ as $n \to \infty$? Explain your answer.

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Q2. (Q3, 24/25)

(a) Define the *natural piecewise cubic spline interpolant* to the function f that is continuous on $[a, b]$ at the $N+1$ knot points $\{x_0, x_1, \dots, x_N\}$.

(b) Suppose that S is a natural cubic spline on $[0, 2]$ with

$$S(x) = \begin{cases} 10x + a(1-x)^2 + x^3, & \text{for} \quad 0 \le x < 1 \\ b(2-x) - c(x+1)^2 + d(x-1)^3, & \text{for} \quad 1 \le x \le 2. \end{cases}$$

Find a, b, c and d.

(c) Suppose that S is the cubic spline interpolant to $f(x) = \ln(x)$ on the $N+1$ equally spaced points $\{x_0 = 1 < x_1 < \cdots < x_N = 4\}$. The error estimate between the function f and the cubic spline interpolant S has an upper bound given by

$$\|f - S\|_\infty := \max_{x_0 \le x \le x_N} |f - S| \le \frac{5h^4}{384} \max_{x_0 \le x \le x_N} |f^{(4)}(x)|,$$

where $h = \dfrac{x_N - x_0}{N}$.
What value of N should one take to ensure that $\|f - S\|_\infty$ is no more than 10^{-6}?

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Q3. (Q4, 24/25) Consider the following differential equation:

$$-u''(x) + u(x) = x \quad \text{on} \quad (0, 1) \quad \text{with} \quad u(0) = u(1) = 0 \tag{2}$$

(a) State the variational formulation of the differential equation (2).
Show that the solution to the variational problem is unique.

(b) Suppose you wanted to compute an approximation to (2) using the finite element method (FEM) on the uniform mesh $\{x_0, x_1, \dots, x_n\}$. Denote the set of the usual piecewise linear Galerkin basis ("hat") functions as $\{\psi_1, \psi_2, \dots \psi_n\}$.

i. Give a formula for a typical basis function ψ_i and give a rough sketch of the basis function.

ii. The FEM applied to (2) leads to a linear system of equations, which can be written as the matrix-vector equation $Ax = b$. Give an expression for the entries of A and b in terms of the basis functions ψ_i. (You do not have to derive an explicit formula for these entries.)

iii. Explain why A is symmetric and tridiagonal.

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Q4. (Q2, 23/24)

(a) Let $\{x_0, x_1, x_2\} := \{1, 4, 9\}$ be a set of interpolation points. Write down the formulae for the associated Lagrange Polynomials, $\{L_0, L_1, L_2\}$, and give a rough sketch of them, clearly indicating their values at the interpolation points. (You can make the sketch in your answer book; graph paper is not needed).

Suppose we define $q(x) := 1 - L_0(x) + L_1(x) + L_2(x)$. Show that, in fact, $q(x) \equiv 0$, for all x.

(b) Let $f(x) = x^{3/2}$. Give the Lagrange form of p_2, the polynomial interpolant to f at $x_0 = 1$, $x_1 = 4$, and $x_2 = 9$.

Using (1), give an upper bound for $|f(7) - p_2(7)|$.

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Q5. (Q3 23/24)

(a) Let l be the piecewise linear interpolant to a given function f on $N+1$ *equally spaced points* in the interval $[a, b]$. Let $h = (b-a)/N$. Use (1) to give an upper bound for $\max\limits_{a \le x \le b} |f(x) - l(x)|$, in terms of h and f''.

(b) Let $f(x) = \ln(x)$, $a = 1$, $b = 3$ and $N = 2$. Write down a formula for its piecewise linear interpolant l.

Give an upper bound for $\max\limits_{a \le x \le b} |f(x) - l(x)|$.

What value of N would you have to choose to ensure that $\max\limits_{a \le x \le b} |f(x) - l(x)| \le 10^{-4}$?

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Q6. (Q4, 23/24)

(a) We denote the Newton-Cotes quadrature rule for approximating $\int_{-1}^{1} f(x)dx$ as

$$Q_N(f) := \sum_{i=0}^{N} q_i f(x_i).$$

Show that $q_i = q_{N-i}$ for $i = 0, 1, \ldots, N$.

(b) Consider the following rule for approximating $\int_0^4 f(x)dx$:

$$R(f) = q_0 f(0) + q_1 f(1) + \frac{2}{9} f(3) + f(4).$$

i. Determine the values of q_0 and q_1 so that $R(\cdot)$ has precision 1.

ii. What is the maximum precision of $R(\cdot)$ for the values of q_0 and q_1 that you have determined?

iii. Why is $R(\cdot)$ not a Newton-Cotes rule, in the strictest sense?